

Introduction

This project analyzed natural hazard risk across six New England states using FEMA National Risk Index (NRI) data and Social Vulnerability Index (SVI) data. The main focus of the project was winter weather risk and how it changes across Maine, New Hampshire, Vermont, Massachusetts, Rhode Island, and Connecticut. The goal was to identify regional trends, compare differences between states, and understand how communities may be affected by winter weather hazards.

The project also looked at how population and community vulnerability may affect the impact of natural disasters. Maps and graphs were used to clearly show patterns and make the information easier to understand.

Methods

The analysis was completed in Python using several data science and visualization libraries, including Pandas, NumPy, Matplotlib, Seaborn, and GeoPandas. FEMA Census Tract datasets and map shapefiles were imported and cleaned before analysis. Invalid values, such as -9999, were removed or replaced to improve the accuracy of the data.

The dataset was filtered to include only the six selected New England states. The winter weather risk variable (WNTW_RISKV) was used to compare hazard levels across census tracts. Geographic map data was also adjusted into a consistent format so accurate maps could be created.

Several maps and graphs were generated to compare winter weather risk between states and identify regional trends. Functions were created in the code to organize repeated tasks such as filtering data and generating visualizations. The project also included discussions about possible bias in the datasets and how coding decisions may affect the final results.

Discussion

The results showed clear differences in winter weather risk across the New England region. Northern states such as Maine, Vermont, and New Hampshire generally had higher winter weather risk values than southern states such as Rhode Island and Connecticut. Massachusetts showed mixed results depending on the area being analyzed.

The maps and graphs showed that geography and climate play a major role in winter weather risk. Areas with harsher winters and more rural communities often showed higher risk levels. Population density and social vulnerability may also affect how strongly communities are impacted during severe winter weather events.

Overall, the project successfully used programming, data cleaning, and mapping tools to analyze natural hazard risk across the region. The visualizations helped explain regional differences in winter weather risk and showed how environmental data can be used to better understand and prepare for natural disasters.

References

- SVI Data & Documentation Download*. (2024, December 16). Place and Health - Geospatial Research, Analysis, and Services Program (GRASP). <https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html>
- Resilience Analysis and Planning Tool (RAPT)*. (2026, April 8). FEMA.gov. <https://www.fema.gov/emergency-managers/practitioners/resilience-analysis-and-planning-tool>